

LGG Brain MRI Tumor Segmentation Using U-Net

Project Report

Dataset: LGG Brain MRI Segmentation Dataset

Model: U-Net

Framework: PyTorch

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1. Introduction

Brain tumor segmentation is an important task in medical image analysis because it aims to identify abnormal tissue at the pixel level in medical images. Accurate segmentation can support quantitative analysis and further clinical or research applications.

In this project, a U-Net convolutional neural network was developed for binary brain tumor segmentation from the LGG (Low-Grade Glioma) MRI dataset. The project focuses not only on model development but also on establishing a reliable experimental pipeline, including image-mask verification, patient-level dataset splitting, preprocessing, model training, quantitative evaluation, and qualitative analysis.

Two different loss-function configurations were investigated:

- **Binary Cross-Entropy (BCE) + Dice Loss**
- **Tversky Loss with $\alpha = 0.3$ and $\beta = 0.7$**

The main objective was to compare the behavior of these loss functions under the same U-Net architecture and dataset split, with particular attention to Dice performance,

precision, recall, and segmentation behavior.

2. Dataset

The project uses the LGG brain MRI segmentation dataset [1] containing 3,929 MRI images and 3,929 corresponding segmentation masks.

Each MRI image has a spatial resolution of 256×256 pixels. The original MRI images are stored as RGB images, while the segmentation masks are grayscale images. The masks represent binary tumor regions.

Dataset Characteristics

Property	Value
MRI images	3,929
Segmentation masks	3,929
Matched image-mask pairs	3,929
Unique patients	110
Image resolution	256×256
Original image format	RGB
Mask format	Grayscale
Empty masks	2,556
Non-empty masks	1,373

The dataset contains a large number of images with empty masks. Specifically, 2,556 masks contain no tumor pixels, while 1,373 contain tumor regions. This distribution creates a challenging segmentation setting because tumor pixels occupy only a relatively small portion of the overall image in many samples.

Image-Mask Pairing

A filename-based matching procedure was used to pair each MRI image with its corresponding mask. Rather than relying on the independent ordering of image and mask file lists, a dictionary was constructed from mask filenames and each image was matched to the corresponding mask.

The pairing was subsequently verified using assertions to ensure that every image-mask pair shared the same base filename. The final pipeline reported 3,929 correctly matched pairs.

This verification step is particularly important in medical image segmentation because an incorrect image-mask association can invalidate the training and evaluation process.

Patient-Level Dataset Split

To reduce the risk of patient-level data leakage, the dataset was divided according to patient IDs rather than individual images.

The resulting split was:

Split	Patients	Image-Mask Pairs
Training	77	2,750
Validation	16	618
Test	17	561
Total	110	3,929

The split was generated using a fixed random seed of 42. The notebook verified that there was no overlap between the patient IDs in the training, validation, and test subsets.

3. Methodology

3.1 Preprocessing

MRI images were converted from RGB to grayscale before being passed to the U-Net model. Pixel intensities were normalized to the range $[0, 1]$ by dividing by 255.

The segmentation masks were also converted to grayscale and transformed from their original representation of 0 and 255 into binary values of 0 and 1.

The resulting tensors have the following structure:

- Input: $1 \times 256 \times 256$
- Target mask: $1 \times 256 \times 256$

The preprocessing pipeline was implemented consistently inside the custom PyTorch dataset class.

3.2 U-Net Architecture

A U-Net architecture was implemented using PyTorch.

The network consists of:

- Four encoder stages
- A bottleneck layer
- Four decoder stages
- Max-pooling for downsampling
- Transposed convolutions for upsampling
- Skip connections between corresponding encoder and decoder levels

- Two convolutional layers with Batch Normalization and ReLU activation at each double-convolution block
- A final 1×1 convolution producing one output channel

The number of feature channels increases from 64 in the first encoder block to 1024 at the bottleneck before being progressively reduced in the decoder. The model takes a single-channel MRI image as input and produces a single-channel segmentation logit map as output.

A pre-training sanity check was also performed to verify that the model accepts a $1 \times 256 \times 256$ input and produces an output with the same spatial resolution.

3.3 Loss Functions

BCE + Dice Loss

The first experiment combines Binary Cross-Entropy with Dice Loss:

Total Loss = BCE Loss + Dice Loss

BCE provides pixel-level supervision, while Dice Loss directly considers the overlap between predicted and ground-truth tumor regions.

The Dice coefficient used in the implementation is defined as:

$$\text{Dice} = (2 \times \sum(\mathbf{P} \times \mathbf{T}) + \epsilon) / (\sum \mathbf{P} + \sum \mathbf{T} + \epsilon)$$

where $\mathbf{P}$ represents the predicted probability map after applying the sigmoid function, $\mathbf{T}$ represents the ground-truth mask, and ϵ is a small smoothing constant used to improve numerical stability.

Tversky Loss

The second experiment uses Tversky Loss, which is defined as:

$$\text{Tversky Loss} = 1 - \text{Tversky Index}$$

where the Tversky Index (TI) is given by:

$$TI = (TP + \epsilon) / (TP + \alpha FP + \beta FN + \epsilon)$$

where **TP**, **FP**, and **FN** represent true positives, false positives, and false negatives, respectively. The parameters α and β control how much false positives and false negatives are penalized, while ϵ is a small smoothing constant used for numerical stability.

In this experiment, $\alpha = 0.3$ and $\beta = 0.7$ were selected, resulting in a higher penalty for false negatives than for false positives. This configuration was chosen to investigate a loss function with a stronger emphasis on missed tumor regions.

Since only one Tversky parameter configuration was evaluated, the results should be interpreted as a comparison of this specific configuration rather than a general evaluation of different Tversky settings.

3.4 Evaluation Metrics

The following metrics were used:

- **Dice Score** — primary segmentation metric
- **Intersection over Union (IoU)**
- **Precision**
- **Recall**

For IoU, Precision, and Recall, model probabilities were converted into binary predictions using a threshold of 0.5.

For the final test evaluation, Dice was calculated as a dataset-level global soft Dice using accumulated prediction probabilities and target pixels, while IoU, Precision, and Recall were calculated from thresholded binary predictions.

The overall workflow of the project is summarized in Figure 1.

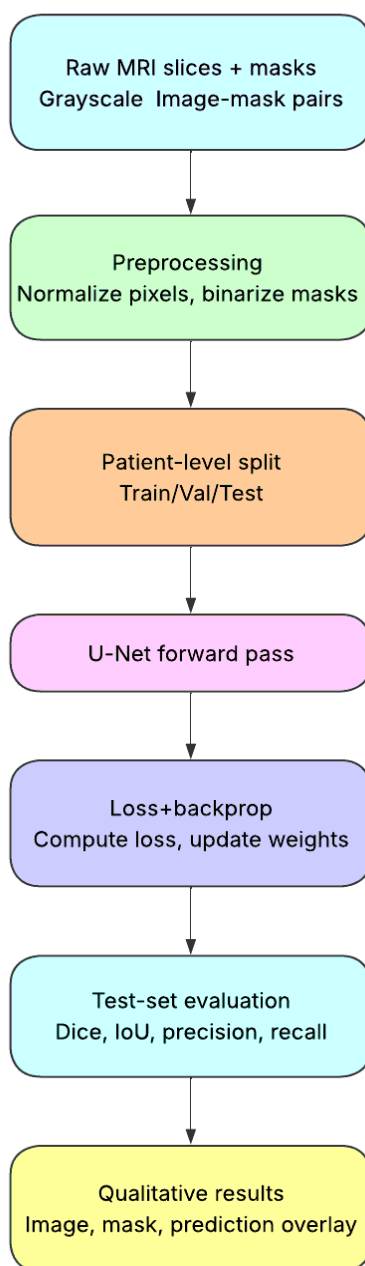


Figure 1. Overall experimental pipeline of the proposed U-Net-based brain tumor segmentation workflow.

4. Experimental Setup

Both experiments used the same:

- Dataset
- Patient-level train/validation/test split
- Image preprocessing
- U-Net architecture
- Optimizer
- Learning rate
- Batch size
- Number of training epochs

Training Configuration

Parameter	Setting
Architecture	U-Net
Input	$1 \times 256 \times 256$
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	16
Maximum epochs	30
Early-stopping patience	7
Random seed	42
Computation device	CUDA-enabled GPU

The experiments were executed on CUDA-enabled GPU hardware. Due to limited and

non-continuous access to suitable GPU computing resources, training was conducted through remote execution and monitoring rather than on a continuously available local GPU system. This constraint limited the number of independent training runs and the extent of hyperparameter experimentation.

The BCE+Dice experiment was trained for all 30 epochs. Its best validation checkpoint was selected according to validation Dice. The training log shows that the best validation Dice of 0.4821 was obtained at epoch 29.

The Tversky experiment also used 30 epochs, a batch size of 16, a learning rate of $1e-4$, and an early-stopping patience of 7 epochs.

5. Experiments

Experiment 1 — BCE + Dice Loss

The first experiment trained the U-Net using the combined BCE and Dice loss.

Training loss decreased significantly throughout the experiment. Validation Dice improved during training but showed fluctuations in the later epochs.

The highest validation Dice observed during this training run was:

Best Validation Dice = 0.4821 at Epoch 29

At epoch 30, validation Dice decreased to 0.4137 while training Dice remained higher at 0.8587. The corresponding training and validation losses were 0.1518 and 0.6312, respectively.

The difference between training and validation behavior in the later epochs indicates that improvements on the training set were not consistently followed by improvements on the validation set. This indicates some degree of reduced generalization in the later stage of training.



Figure 2. Training and validation loss during the BCE + Dice experiment.

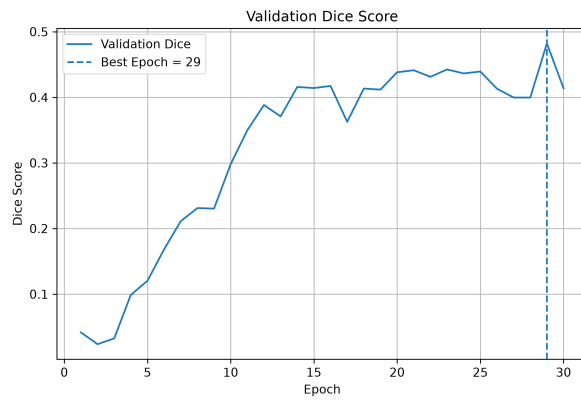


Figure 3. Validation Dice score during the BCE + Dice experiment.

The epoch-29 value is reported as the best observed validation performance in this training run. The decrease at epoch 30 indicates late-stage validation fluctuation, but a single training run is insufficient to determine whether the epoch-29 peak represents random noise or a better model state.

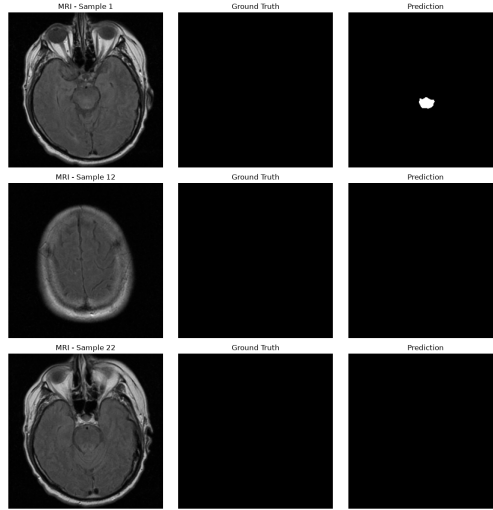


Figure 4. Qualitative segmentation examples produced by the BCE + Dice model, showing the MRI input, ground-truth tumor mask, and predicted segmentation.

Experiment 2 — Tversky Loss

The second experiment trained a U-Net using Tversky Loss with $\alpha = 0.3$ and $\beta = 0.7$.

The highest validation Dice was:

Best Validation Dice = 0.2573 at Epoch 25

The validation Dice fluctuated and reached 0.2533 at epoch 30. The model was therefore not able to improve upon its epoch-25 validation performance.

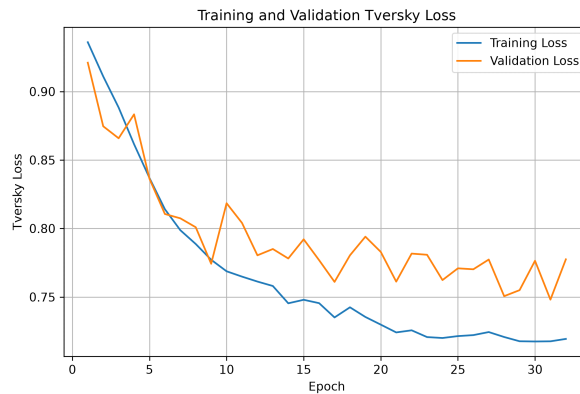


Figure 5. Training and validation loss during the Tversky loss experiment ($\alpha = 0.3$, $\beta = 0.7$).

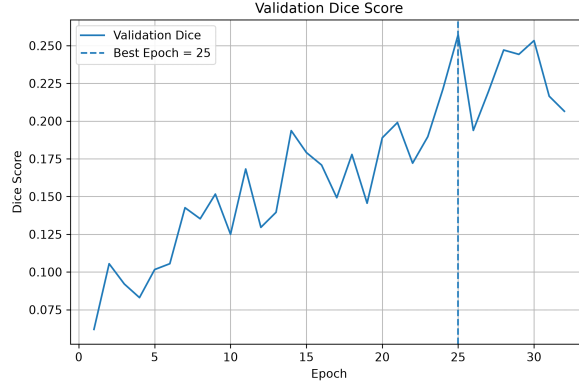


Figure 6. Validation Dice score during the Tversky loss experiment ($\alpha = 0.3$, $\beta = 0.7$).

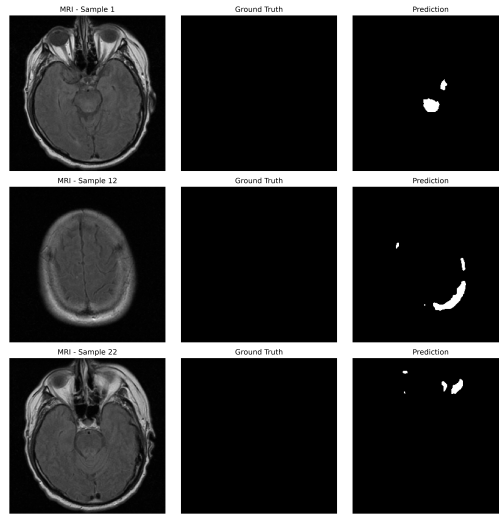


Figure 7. Qualitative segmentation examples produced by the Tversky model ($\alpha = 0.3$, $\beta = 0.7$), showing the MRI input, ground-truth mask, and predicted segmentation.

6. Results

Quantitative Results

Experiment	Best Val Dice	Best Epoch	Test Dice	Test IoU	Test Precision	Test Recall
BCE + Dice	0.4821	29	0.7044	0.5492	0.7557	0.6678
Tversky ($\alpha=0.3$, $\beta=0.7$)	0.2573	25	0.2476	0.1551	0.1681	0.6667

The BCE+Dice experiment achieved higher test performance under the selected

experimental settings. Its test Dice was 0.7044, with an IoU of 0.5492, Precision of 0.7557, and Recall of 0.6678.

The Tversky experiment achieved a test Dice of 0.2476, IoU of 0.1551, Precision of 0.1681, and Recall of 0.6667. These values were saved as the final Tversky results in the notebook.

Training Behavior

The BCE+Dice model demonstrated considerably stronger learning behavior, with training Dice increasing to 0.8587 by epoch 30. However, validation performance did not follow the same trend, indicating a gap between training and validation behavior.

The Tversky model showed much slower improvement in training Dice and lower validation Dice. Its relatively high recall compared with its precision indicates that the selected configuration did not achieve a good balance between detecting tumor pixels and avoiding false-positive predictions.

7. Discussion

Under the experimental conditions used in this project, BCE + Dice achieved better overall segmentation performance than the tested Tversky configuration.

The difference was particularly evident in Precision and Dice. BCE+Dice achieved a test Precision of 0.7557, whereas Tversky achieved 0.1681. At the same time, both experiments produced similar Recall values, approximately 0.67.

This pattern suggests that the Tversky configuration with $\alpha = 0.3$ and $\beta = 0.7$ was able to maintain sensitivity to tumor pixels but produced considerably more false-positive predictions. This finding matches the formulation of the loss, in which false negatives receive greater weighting than false positives.

A visual assessment also showed cases in which the Tversky model predicted tumor regions where the ground-truth mask was empty. This behavior is consistent with a

tendency toward over-segmentation and false-positive predictions.

However, these findings should not be interpreted as evidence that Tversky Loss is worse than BCE+Dice. Only one pair of Tversky parameters was tested, and different α/β values could lead to different behavior.

The BCE+Dice experiment also showed limitations. Although the test performance was higher, the training and validation curves indicate that the model's ability to generalize did not improve during the final epochs. In particular, the best validation Dice occurred at epoch 29, followed by a noticeable decrease at epoch 30. This supports selecting the best validation checkpoint rather than automatically using the final epoch. However, because only one run was performed, the stability of the epoch-29 peak cannot be established without repeated experiments.

An additional methodological strength of the project is that image-mask correspondence and patient-level separation were carefully validated. These steps reduce the risk of silent data errors and patient-level leakage and make the experimental comparison more reliable.

8. Limitations

Several limitations should be considered when interpreting the results.

8.1 Limited Hyperparameter Search

The experiments used a fixed learning rate, batch size, number of training epochs, and U-Net configuration. An additional hyperparameter search was not performed.

8.2 Single Tversky Configuration

Only $\alpha = 0.3$ and $\beta = 0.7$ were evaluated. Therefore, the observed Tversky behavior cannot be generalized to all Tversky configurations.

8.3 Limited and Non-Continuous GPU Access

Although the final experiments were executed on CUDA-enabled GPU hardware, suitable GPU resources were not continuously available to the researcher. Training was therefore performed through remote execution and monitoring.

This limitation reduced the number of independent training runs and the amount of hyperparameter experimentation. Consequently, the study focused on two controlled loss-function experiments rather than a detailed optimization study.

8.4 Limited Training Budget

Training was limited to 30 epochs. Although the BCE+Dice model continued to show clear improvement in the training set, validation performance fluctuated during the later epochs. Therefore, simply increasing the number of epochs would not necessarily improve generalization and could potentially increase overfitting.

8.5 Class Imbalance

A large proportion of the dataset contains empty masks. In addition, tumor regions can occupy relatively small areas of the images. This creates an imbalanced segmentation problem that can make optimization more difficult.

8.6 Limited Data Augmentation

The current pipeline does not include an extensive augmentation strategy. Additional augmentation could potentially improve robustness to variations in appearance and spatial characteristics.

8.7 Single Experimental Run

Each loss configuration was evaluated using one training run with a fixed random seed. Multiple independent runs would provide a better estimate of variability and experimental

stability.

9. Future Work

Several improvements could be investigated in future experiments:

- **Tversky parameter tuning** — Evaluate multiple α/β combinations to determine whether a better balance between false positives and false negatives can be achieved.
- **Data augmentation** — Introduce appropriate spatial and intensity augmentations for MRI segmentation.
- **Learning-rate scheduling** — Evaluate learning-rate schedulers to improve optimization and validation performance.
- **Regularization** — Investigate dropout, weight decay, or other regularization techniques to reduce overfitting.
- **Alternative loss functions** — Compare BCE+Dice and Tversky with other segmentation losses such as Focal Loss, Focal Tversky, or combinations designed for highly imbalanced segmentation.
- **Longer and controlled training** — Use a larger computational budget together with early stopping and best-checkpoint selection rather than simply increasing the number of epochs.
- **Multiple experimental runs** — Repeat each experiment with different random seeds to evaluate result stability.
- **Detailed error analysis** — Perform separate analyses for images with and without tumors and examine false-positive and false-negative cases at the patient level.

- **Stronger architectures** — Future work could investigate U-Net variants, pretrained encoders, attention mechanisms, or other medical image segmentation architectures.

10. Conclusion

This project developed a complete U-Net-based pipeline for binary brain tumor segmentation using LGG MRI data.

The workflow included image-mask pairing verification, patient-level dataset splitting, grayscale preprocessing, binary mask conversion, U-Net training, checkpointing, quantitative evaluation, and qualitative analysis.

Two loss-function configurations were investigated. Under the experimental settings used in this study, BCE + Dice demonstrated better segmentation performance than the tested Tversky configuration, achieving a best validation Dice of 0.4821 and a test Dice of 0.7044, compared with 0.2573 and 0.2476 for Tversky, respectively.

The Tversky experiment nevertheless provided useful information about the effect of emphasizing false negatives. Its recall was similar to BCE + Dice, but its much lower precision indicated a higher rate of false-positive tumor predictions.

Overall, the project should be considered a baseline experimental study rather than a fully optimized segmentation benchmark. Its main value lies not only in the final numerical results but also in the structured research workflow, data-integrity checks, controlled loss-function comparison, error analysis, and discussion of limitations and future improvements.

11. References

[1] Buda, M., Saha, A., & Mazurowski, M. A. (2019). Association of genomic subtypes of lower-grade gliomas with shape features automatically extracted by a deep learning algorithm. *Computers in Biology and Medicine*, 109, 318–328.

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